

Marginally Useful: Conformal Prediction Adds No Forecastability

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Abstract

The conformal step adds no forecastability. A conformal predictive system pools one law for its score across all inputs, and its expected log-score regret to the oracle is exactly the mutual information between that score and the input, plus a marginal mismatch the conformal step removes. The information the system can extract about the future is therefore fixed by the model and score it wraps before calibration runs, and it can be less than none: a poorly located model wrapped in conformal prediction can do worse than ignoring the inputs altogether. On 572 economic time series, pooled raw residuals leave at least 0.117 nats per observation to a feasible challenger, and a single scale transform leaves none it can detect.

1 The claim

Conformal prediction is widely presented as the rigorous way to add uncertainty to a forecast. One forecasting library tabulates methods by “Calibration Guarantee”, awarding the check mark to conformal prediction and a tilde to bootstrap, quantile regression and ARIMA alike (Nixtla, 2026). A popular introduction adds that with Bayesian posteriors or bootstrapping “we have no guarantees that these approaches will perform well on new data” (Molnar, 2023), and from there it is a short step to conformal prediction as the thing that “converts any statistical, machine or deep learning model into a probabilistic prediction model” (Manokhin, 2023). The methodological literature is more careful (Lei and Wasserman, 2014; Foygel Barber et al., 2021), but it states its cautions in the currency of coverage. The question a forecaster asks is in a different currency: how much of what can be predicted does the method predict?

That question has a precise form. Under the logarithmic score the most any forecast can gain over the unconditional distribution of Y is the mutual information $I(Y; X)$ between the outcome and the information set (DelSole, 2004), and Catt (2026) makes this the definition of *forecastability*, resolved by horizon. We answer the question for conformal prediction. A conformal predictive system captures $I(Y; X) - I(T; X)$ of the available forecastability, where T is the score it pools, and the conformal step contributes none of it (Theorem 1 and Corollary 1). The share can be negative. Whatever a conformal system knows about the future, the model and score it wraps knew before calibration ran.

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None of this disputes the guarantee. Given exchangeable data, a fitted predictor and a nonconformity score, split conformal prediction returns a set with

$$\mathbb{P}(Y_{n+1} \in C_\alpha(X_{n+1})) \geq 1 - \alpha, \quad (1)$$

finite-sample and with no assumption on the data distribution (Vovk et al., 2005; Lei et al., 2018; Angelopoulos and Bates, 2023). That is a statement about coverage, averaged over X_{n+1} . Our result is about the log score, and it separates cleanly from (1): the finite-sample guarantee is coverage, the log-score statement is exact for every fixed pooled law and asymptotic only in identifying which law the conformal system uses (Lemma 1).

2 Setup

Let (X, Y) have joint law P on $\mathcal{X} \times \mathbb{R}$, with conditional density $p(y | x)$ and marginal density p_Y . In time series, X is the past and Y the value h steps ahead. The oracle forecasts with $p(\cdot | x)$ and the unconditional forecaster with p_Y . Following Catt (2026), the forecastability of Y given X is

$$F = \mathbb{E} \log p(Y | X) - \mathbb{E} \log p_Y(Y) = I(Y; X), \quad (2)$$

the most any forecast can gain over the unconditional law in expected log score. For a forecaster q , its *exploitability* is $\mathbb{E} \log q(Y | X) - \mathbb{E} \log p_Y(Y)$, which is at most F and can be negative.

Scores and pooling. A score transform is a family of maps $\phi_x : \mathbb{R} \rightarrow \mathcal{Z}$, one per input, each a strictly increasing C^1 bijection onto a common open interval $\mathcal{Z}$, with $(x, y) \mapsto \phi_x(y)$ and $(x, y) \mapsto \partial_y \phi_x(y)$ jointly measurable. Write $T = \phi_X(Y)$ for the score and P_T, p_T for its marginal law and density. The cases in use are

- the signed residual $\phi_x(y) = y - \hat{\mu}(x)$, whose conformal predictive system (CPS) is that of Vovk et al. (2019);
- the normalized residual $\phi_x(y) = (y - \hat{\mu}(x)) / \hat{\sigma}(x)$ (Lei et al., 2018);
- the estimated conditional rank $\phi_x(y) = \Phi^{-1}(\hat{F}(y | x))$, which is distributional conformal prediction (Chernozhukov et al., 2021).

The model $(\hat{\mu}, \hat{\sigma}$ or $\hat{F})$ is fitted on data disjoint from what follows and is held fixed. A *pooled forecaster* in the coordinates ϕ chooses one density h on $\mathcal{Z}$, the same for every input, and reports

$$q_h(y | x) = h(\phi_x(y)) \partial_y \phi_x(y). \quad (3)$$

A conformal predictive system is a pooled forecaster whose h is read off the calibration scores. Conformal calibration sees the scores T_i and never the inputs, so it can choose h and nothing else.

3 The gap

Theorem 1 (Information gap of a pooled forecaster). *Assume T has finite differential entropy, $I(T; X) < \infty$, and $\mathbb{E} |\log h(T)| < \infty$. Then for every density h on $\mathcal{Z}$,*

$$\mathbb{E} \log p(Y | X) - \mathbb{E} \log q_h(Y | X) = I(T; X) + \text{KL}(P_T \| H). \quad (4)$$

The regret is minimized at $h = p_T$, where it equals $I(T; X)$. No choice of h , and no recalibration that ignores X , brings it below $I(T; X)$.

Proof. By the change of variables $t = \phi_x(y)$, $p(y|x) = p_T(\phi_x(y)|x) \partial_y \phi_x(y)$, where $p_T(\cdot|x)$ is the conditional density of T . The Jacobians cancel in the ratio with (3), so

$$\mathbb{E} \log \frac{p(Y|X)}{q_h(Y|X)} = \mathbb{E} \log \frac{p_T(T|X)}{h(T)} = \mathbb{E} \log \frac{p_T(T|X)}{p_T(T)} + \mathbb{E} \log \frac{p_T(T)}{h(T)} = I(T; X) + \text{KL}(P_T \| H),$$

each term finite under the assumptions. The second term is non-negative and vanishes only at $h = p_T$ (Gibbs). An X -blind recalibration applies a fixed increasing map ψ to the score, and $I(\psi(T); X) = I(T; X)$ because ψ is invertible, so the floor is unchanged. $\square$

The identity (4) is the chain rule for relative entropy along a product reference, the split of redundancy into a dependence term and a marginal-mismatch term familiar from universal coding (Clarke and Barron, 1990; Grünwald, 2007). We claim the reading, not the algebra: it holds exactly, at every sample size, for the whole class of pooled forecasters, which is the class every conformal predictive system belongs to. For the signed residual it gives the residual gap $I(R; X)$ with $R = Y - \hat{\mu}(X)$. The form in general coordinates is also stated by Cotton (2026), who use it to measure the gap along a sequence of transforms. Reading the symmetric intervals of ordinary absolute-residual conformal as a distribution pools the symmetrized law $h_{\text{sym}}(z) = \frac{1}{2}(p_R(z) + p_R(-z))$, and (4) then adds $\text{KL}(P_R \| H_{\text{sym}}) \leq \log 2$ for discarding the sign.

Lemma 1 (Which law the conformal system pools). *Let the calibration pairs (X_i, Y_i) , $i = 1, \dots, n$, be i.i.d. copies of (X, Y) , or a stationary ergodic sequence with that marginal law, and let $\hat{G}_n$ be the empirical distribution function of the scores $T_i = \phi_{X_i}(Y_i)$. The signed CPS reports the predictive distribution function $\Pi_x(y) = \hat{G}_n(\phi_x(y))$, up to the $1/(n+1)$ fence-post convention, and $\sup_t |\hat{G}_n(t) - G_T(t)| \rightarrow 0$ almost surely, by Glivenko–Cantelli in the i.i.d. case and by the ergodic theorem applied on a countable dense set of t in the stationary case. If a smoothed version with density $\hat{h}_n$ satisfies $\text{KL}(P_T \| \hat{H}_n) \rightarrow 0$, the regret (4) of the smoothed CPS tends to $I(T; X)$.*

Exchangeability alone does not give this. An infinite exchangeable sequence is a mixture of i.i.d. sequences, and $\hat{G}_n$ then converges to the distribution function of the directing measure, which is random. Coverage (1) needs only exchangeability; the identification needs more. A finite empirical CPS is a predictive distribution but not a Lebesgue density, which is why the log-score statement is made for a smoothed version. Theorem 1 needs none of this, because it is a statement about every fixed h .

Corollary 1 (What a conformal system captures). *Under the assumptions of Theorem 1, and with $I(Y; X) < \infty$, the pooled forecaster (3) has exploitability*

$$\mathbb{E} \log q_h(Y|X) - \mathbb{E} \log p_Y(Y) = I(Y; X) - I(T; X) - \text{KL}(P_T \| H). \quad (5)$$

Its best value, $I(Y; X) - I(T; X)$, is fixed by the score before calibration. The fraction of forecastability captured is at most $1 - I(T; X)/I(Y; X)$, which is negative whenever $I(T; X) > I(Y; X)$.

Proof. Subtract (4) from (2). $\square$

Remark 1 (Why the calibration gain is not forecastability). The conformal step moves h toward p_T and so raises the score by removing $\text{KL}(P_T \| H)$. That term is a property of the marginal law of T alone. It is the same whether X carries a great deal of information about Y or none, and a forecaster with $I(Y; X) = 0$ gains it just as much. What depends on X is $I(Y; X) - I(T; X)$, and that is set by ϕ . In this sense the conformal step adds no forecastability: it repairs marginal miscalibration, and what the system knows about the future is decided upstream.

Remark 2 (Negative capture). The corollary’s negative case is not exotic. Let Y be independent of X , so there is nothing to forecast, and let $\hat{\mu}$ vary with x . Then $R = Y - \hat{\mu}(X)$ depends on X through the model alone, $I(R; X) > 0 = I(Y; X)$, and the conformal predictive system, perfectly calibrated in coverage, scores below the unconditional law by $I(R; X)$. The coverage certificate is silent about this.

Remark 3 (Cells and conditional methods). Mondrian or binned conformal predictive systems (Boström et al., 2021; Toccaceli, 2026) pool one law per cell $g(X)$. The same computation with H replaced by a family H_g gives the floor $I(T; X | g(X)) = I(T; X) - I(T; g(X))$: calibration per cell captures $I(T; g(X))$ and no more. Distributional conformal prediction (Chernozhukov et al., 2021) is Theorem 1 with $\phi_x = \Phi^{-1} \circ \hat{F}(\cdot | x)$, so its floor is $I(T; X)$ for the estimated rank, which is zero when $\hat{F}$ is the true conditional law. That is the sense in which it answers the problem posed here, and it does so through $\hat{F}$. Density-based and conditional-histogram constructions (Izbicki et al., 2020, 2022; Sesia and Romano, 2021), localized conformal prediction (Guan, 2023) and conformalized quantile regression (Romano et al., 2019) all put X into the score or the reference class. Each reduces the floor by what its conditioning retains, never by the calibration step.

4 Two readings

A false-pooling cost. A pooled forecaster asserts that once ϕ is applied, one law fits every input, which is to say $T \perp X$. The gap $I(T; X)$ is the relative entropy from the true joint law to the nearest such independence model, and one pays it whether or not one notices making the assumption.

A betting rent. A gambler who knows a distribution P and bets against a fair book priced by Q compounds wealth at expected log-rate $\text{KL}(P \| Q)$ per round (Kelly, 1956; Cover and Thomas, 2006). Read the pooled law as the book and the conditional law as what the oracle knows; the oracle’s expected rent per round is $I(T; X)$, and a re-levelling that ignores X leaves it unchanged.

Coverage and log score grade different objects. Two forecasters with the same location $\hat{\mu}$ but different claimed densities f have identical residual conformal sets and identical coverage at every level, while their log scores differ without bound: take X degenerate, $Y \sim N(0, 1)$ and $f_\sigma = N(0, \sigma^2)$, whose expected log score $-\frac{1}{2} \log(2\pi\sigma^2) - 1/(2\sigma^2)$ diverges at both ends of the σ range while the set never moves. This does not conflict with Theorem 1, where the degenerate X gives $I(T; X) = 0$. The example grades a density f that the conformal set never consults; the theorem grades the density the conformal system itself reports. If the practitioner replaces a misspecified f by the conformal system’s output, the score can rise, by the KL term of Remark 1, and never past the ceiling $I(Y; X) - I(T; X)$.

5 Evidence

Exact checks. The identities are checked by deterministic grid integration, each side computed independently from its definition (`check_gap.py`). With X uniform on $[0, 1]$ and $Y | x$ skew-normal with mean $2 \sin 3x$ and scale $0.5 + 1.5x$:

location model $\hat{\mu}(x)$	$I(Y; X)$	$I(R; X)$	captured	share
correct, $2 \sin 3x$	0.340	0.118	0.222	0.65
shrunk, $1.2 \sin 3x$	0.340	0.152	0.189	0.56
$Y \perp X$, $\hat{\mu}(x) = 1.5 \cos 4x$	0	0.653	-0.653	n/a

Both sides of (4) and (5) agree to within 10^{-8} nats. With the same truth and a misspecified logistic $\hat{F}$, the distributional-conformal score has $I(T; X) = 0.070$ against $I(R; X) = 0.123$ for raw residuals under the same location: the score transform, not the calibration, halves the gap.

Economic time series. Cotton (2026) measures the residual gap on 572 continuous FRED change series, with a strong online forecaster’s mean held fixed and the past as X . The oracle is unavailable, but the gap is still observable from below. If M is the pooled law estimated from the past and Q any past-measurable challenger, then $\mathbb{E} \log(q/m) = \mathbb{E} \text{KL}(P \parallel M) - \mathbb{E} \text{KL}(P \parallel Q) \leq \mathbb{E} \text{KL}(P \parallel M)$, so a challenger’s held-out gain over the pooled law is, in expectation, a lower bound on the regret the pooled system pays to the oracle. The held-out gains, in nats per observation, were:

pooled stream	challenger gain	90% interval	positive on
raw residuals	+0.117	[0.105, 0.129]	84% of series
standardized by an EWMA scale	-0.020	[-0.060, 0.015]	42%
serially whitened	-0.016	[-0.058, 0.017]	40%
PIT-normalized	+0.003	[-0.006, 0.012]	26%

Pooling raw residuals, which is what split conformal does, leaves at least 0.117 nats per observation on the table. One scale transform removes all of it that this challenger can find. The calibration step is the same in every row; the rows differ only in the coordinates in which the pooling happens. In a companion simulation with a known heteroscedastic truth, raw and normalized split conformal both hit 90% marginal coverage, while raw pooling covers 99.8% of the calmest volatility quartile and 76.7% of the most volatile (Cotton, 2026).

6 What this does not say

The guarantee stands. Conformal prediction certifies the coverage of a set, finite-sample and distribution-free, under exchangeability. Where coverage is the deliverable, as in selective prediction, shortlisting, or conformal p -values for novelty detection (Angelopoulos et al., 2021; Bates et al., 2021), it is the right tool, and within its pooled class the CPS is log-score optimal. The cost is the class.

Terms. Marginal coverage is one calibration-type property of a set. It is not probabilistic calibration of a predictive distribution, and neither is sharpness (Gneiting et al., 2007; Gneiting and Raftery, 2007). We use *coverage* for the first, *calibration* only for the distributional property, and grade forecasts by the log score.

Heteroscedasticity, not covariate density. The gap is driven by how the law of T varies with x . It has nothing to do with whether x is typical or unusual under p_X : a rare input with ordinary noise contributes nothing to $I(T; X)$, and marginal coverage alone does not say which inputs are over- or under-covered.

The log score. The exact information identities belong to the log score. For CRPS or the interval score the analogous floor is the regret of the best X -blind law under that score, which is not an information quantity, and we claim no identity for it.

Estimate first, certify second. A Bayesian posterior predictive, or any rich conditional model, can supply the distribution, with conformal calibration added on top as a certificate. The theorem says what that order buys: the certificate costs nothing in forecastability if the model has already made T nearly independent of X , and cannot recover what the model left in $I(T; X)$. Conformalize last, and put the conditioning in the score.

Related work. The pieces are known: the CPS (Vovk et al., 2019), the calibration–sharpness paradigm (Gneiting et al., 2007), the impossibility of distribution-free conditional coverage (Lei and Wasserman, 2014; Foygel Barber et al., 2021), an information-theoretic reading of set size (Correia et al., 2024), the observation that validity invites weak predictors (Recht, 2024) and that set length can be gamed at fixed coverage (Min et al., 2026), and forecastability as mutual information (DelSole, 2004; Catt, 2026). To our knowledge the exact regret of the pooled class, its identification with the conformal predictive system, and the resulting account of how much forecastability conformal prediction captures have not been stated before.

7 Conclusion

Coverage answers the question *is the truth in this region?* In forecasting that is often not the question anyone was asking. The question is how much of what can be predicted has been predicted, and for conformal prediction the answer is exact: $I(Y; X) - I(T; X)$ at best, set by the model and the score, with the conformal step contributing nothing that depends on the inputs. The price of the split conformal coverage guarantee, paid in the forecaster’s own currency, is $I(T; X)$.

Data availability. No new data are used. The numerical checks are in `check_gap.py`, supplied with the submission and at the repository above; the time-series results are those of Cotton (2026), whose code and series list are public.

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